TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr.-Ing. Roland Klein	Email	Roland.klein@lbf.fraunhofer.de
Institute Name	Fraunhofer Institute for Structural Durability and System Reliability, LBF			
Address	Schlossgartenstr. 6, 64289 Darmstadt, Germany			

2. Technology Overview

Title of Proposed Technology	**Bio-Based Phenolic Antioxidants as Sustainable Alternatives to 6PPD in Rubber Applications**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (x), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Clean Production	Hazardous Raw/Sub-Materials Substitution Technologies	Fine Chemicals	Additives
Concept of Proposed Technology	The proposed technology centers on the development and implementation of bio-based phenolic antioxidants derived from renewable sources as sustainable replacements for N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylenediamine (6PPD) in rubber compounds, particularly for tire applications. Our innovative approach utilizes natural phenolic substances as building blocks to synthesize multifunctional antioxidants that provide both antioxidant and antiozonant properties. Our technology involves the chemical modification and derivatization of bio-based phenolic compounds to enhance their molecular weight, thermal stability, and compatibility with rubber matrices. The synthesized compounds are designed to interrupt the autoxidation process in rubber through hydrogen donation and radical scavenging mechanisms, while simultaneously providing protection against ozone-induced cracking.			

(max 10 pages)

	Main Functions: - Primary antioxidant activity through phenolic hydrogen donation - Antiozonant protection for unsaturated rubber compounds - Enhanced thermal stability for high-temperature processing - Reduced environmental impact through biodegradable transformation products - Maintained mechanical properties and durability of rubber compounds **Potential Uses:** - Tire tread compounds for passenger and commercial vehicles - Sidewall formulations requiring ozone resistance - Technical rubber products exposed to oxidative environments - Sustainable rubber compounds for environmentally conscious applications.
Rationale for Joint R&D Project	The urgent need for this joint R&D project stems from the growing environmental and regulatory concerns surrounding 6PPD and its toxic transformation product, 6PPD-quinone (6PPD-Q). Recent studies have demonstrated that 6PPD-Q, formed when 6PPD reacts with ozone, exhibits acute toxicity to certain fish species at extremely low concentrations and has been detected in urban runoff, surface waters, and even human samples. **Environmental Drivers:** - 6PPD-Q has been identified as a major cause of coho salmon mortality in urban streams - Regulatory pressure is mounting globally for alternatives to 6PPD in rubber formulations - The tire industry faces increasing scrutiny regarding environmental impact of tire wear particles - Consumer demand for sustainable and environmentally friendly products is growing **Technical Challenges:** 6PPD's unique dual functionality as both antioxidant and antiozonant makes it exceptionally difficult to replace. Most alternative compounds provide only one of these protective mechanisms, requiring complex additive packages that may compromise performance or increase costs. **Collaboration Necessity:** This project requires collaboration because: - **Research Institute Expertise**: Advanced synthetic chemistry, analytical capabilities, and fundamental understanding of antioxidant mechanisms - **Industrial Application Knowledge**: Real-world performance requirements, processing conditions, and market-specific formulation needs - **Scale-up Challenges**: Transition from laboratory synthesis to industrial production requires specialized knowledge and infrastructure - **Validation Requirements**: Comprehensive testing under actual service conditions can only be achieved through industrial partnerships. The Korean rubber industry's leadership in sustainable development and advanced rubber technology makes Korean SMEs ideal partners for this transformative project.
Key Words	1. Bio-based antioxidants 2. Bio-based antiozonants 3. 6PPD replacement 4. Sustainable rubber technology 5. Environmental toxicity reduction

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (7-8)
R&D Objectives	**Primary Objective:** Develop and validate bio-based phenolic antioxidants that can partially or completely replace 6PPD in tire rubber formulations while maintaining or improving performance characteristics and significantly reducing environmental impact. **Specific Technical Objectives:** 1. **Synthesis and Optimization (Phase 1 & 2)** ○ Synthesize at least 5 different derivatives of natural phenolic substances with varying molecular weights and functionalities ○ Optimize synthetic routes for scalability and cost-effectiveness ○ Achieve >95% purity of target compounds through refined purification processes 2. **Performance Validation (Phase 1 & 2)** ○ Demonstrate antioxidant activity equivalent to or better than 6PPD in standardized tests ○ Achieve antiozonant protection within 80-120% of 6PPD performance levels ○ Maintain rubber compound processability and mechanical properties 3. **Environmental Impact Reduction (Phase 2)** ○ Reduce aquatic toxicity by >90% compared to 6PPD-Q transformation products ○ Demonstrate enhanced biodegradability of transformation products ○ Achieve carbon footprint reduction of >30% compared to conventional 6PPD production 4. **Industrial Implementation (Phase 2)** ○ Scale synthesis to pilot plant level (>1 kg batches) ○ Validate performance in real rubber compounds under industrial processing conditions ○ Develop cost-competitive production processes 5. **Knowledge Generation** ○ Establish structure-activity relationships for bio-based phenolic antioxidants in rubber ○ Develop predictive models for antioxidant performance ○ Create intellectual property portfolio through patents and publications			
R&D Target Specifications	**Chemical and Physical Properties:** - Molecular weight: 300-800 g/mol (optimized for low volatility and good dispersion) - Thermal stability: >250°C (suitable for rubber processing temperatures) - Solubility in rubber: Compatible with NR, SBR, and BR compounds - Purity: >95% active ingredient content **Antioxidant Performance Targets:** - Oxidation Induction Time (OIT) at 200°C: ≥80% of 6PPD performance - Oxygen uptake rate reduction: ≥75% compared to unstabilized rubber - Long-term aging resistance: Maintain >90% of initial tensile strength after 168h at 100°C **Antiozonant Performance Targets:** - Ozone resistance (40°C, 50 pphm O_3, 20% strain): No cracking for ≥100 hours - Dynamic ozone test: Performance within 80-120% of 6PPD benchmark - Flex-cracking resistance: ≥1 million cycles without failure **Environmental Performance Targets:** - Aquatic toxicity (LC50, 96h): >100 mg/L for transformation products - Biodegradability: >60% mineralization within 28 days (OECD 301) - Bioaccumulation potential: Log Kow <4.0 for parent compounds **Processing and Economic Targets:** - Processing temperature stability: No degradation up to 180°C - Dispersion quality: Uniform distribution in rubber matrix - Cost target: Within 150% of current 6PPD pricing - Scale-up feasibility: Demonstrated at >1 kg batch size			

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	Regulatory Compliance: - REACH registration pathway identified - Toxicological profile established through preliminary studies - Food contact compliance for relevant applications
R&D Plan	**Phase 1: Proof of Concept and Detailed Planning (6 months)** **Month 1-2: Literature Review and Synthesis Planning** - Comprehensive analysis of bio-based phenolic antioxidants literature - Selection of optimal phenolic derivative structures - Design of synthetic routes and reaction optimization strategies - Procurement of starting materials and analytical standards **Month 3-4: Initial Synthesis and Characterization** - Synthesis of 3-5 target compounds in 10-50g quantities - Structural confirmation using NMR, FTIR, and mass spectrometry - Thermal analysis (TGA, DSC) for processing stability assessment - Preliminary antioxidant activity screening using DPPH and OIT methods **Month 5-6: Preliminary Rubber Testing and Phase 2 Planning** - Incorporation of synthesized compounds into model rubber formulations - Basic mechanical property evaluation (tensile strength, elongation) - Preliminary ozone resistance testing - Detailed Phase 2 project plan development including milestone definitions - Risk assessment and mitigation strategy formulation **Phase 1 Deliverables:** - 3-5 characterized bio-based antioxidant compounds - Preliminary performance data package - Detailed Phase 2 implementation plan - Intellectual property landscape analysis **Phase 2: Development and Validation (30 months)** **Months 7-15: Synthesis Optimization and Scale-up** - Optimization of synthetic routes for improved yield and purity - Scale-up to 100-500g batches - Process safety evaluation and hazard analysis - Development of purification and quality control procedures **Months 7-24: Comprehensive Performance Evaluation** - Systematic antioxidant testing in various rubber compounds - Ozone resistance evaluation under multiple test conditions - Accelerated aging studies (heat, oxygen, ozone exposure) - Mechanical property retention assessment **Months 22-27: Industrial Validation and Application Development** - Collaboration with Korean SME partner for real-world testing - Rubber compound formulation optimization - Processing trials under industrial conditions - Performance validation in demonstrator applications **Months 19-27: Scale-up and Commercialization Preparation** - Pilot-scale synthesis (>1 kg batches) - Comprehensive toxicological and environmental assessment - Regulatory documentation preparation - Cost analysis and market entry strategy development **Months 25-36: Technology Transfer and Implementation** - Technology transfer to Korean SME partner - Joint patent applications and IP protection - Preparation of commercial production processes

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- Market introduction planning and customer validation

Key Methodologies:
- **Synthetic Chemistry**: Multi-step organic synthesis, purification, and characterization
- **Analytical Chemistry**: Advanced spectroscopic and chromatographic methods
- **Rubber Testing**: Standard ASTM and ISO test methods for rubber compounds
- **Environmental Assessment**: Biodegradability, toxicity, and lifecycle analysis
- **Scale-up Engineering**: Process optimization and technology transfer protocols

		Year 1				Year 2				Year 3			
Work package		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
1	Evaluation												
2	Phenolic derivatives												
3	Rubber formulation												
4	Upscaling of phenolic derivatives												
5	Compounding												
6	Transfer to SME												
7	Creation of demonstrator												
Milestones													
1	Production of phenolic derivatives												
2	Performance analysis												
3	Formulation & recipe development												

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Fraunhofer LBF Responsibilities and Contributions:** **Technical Expertise and Research Scope:** - **Synthetic Chemistry Leadership**: Design and execute all synthetic routes for bio-based phenolic antioxidants, leveraging our established expertise in phenolic compound derivative chemistry - **Advanced Analytical Capabilities**: Provide comprehensive characterization using state-of-the-art NMR, FTIR, GC-MS, and thermal analysis equipment - **Rubber Compounding Expertise**: Formulate and process rubber compounds incorporating new antioxidants using our specialized mixing and processing equipment - **Performance Testing**: Conduct systematic evaluation of antioxidant and antiozonant properties using standardized test methods (OIT, ozone resistance, aging studies) **Resources to be Provided:** - **Laboratory Infrastructure**: Synthetic chemistry laboratories, rubber processing equipment (internal mixers, mills), environmental chambers for aging studies - **Analytical Equipment**: High-field NMR spectrometers, FTIR, GC-MS, thermal analysis (TGA, DSC), mechanical testing equipment **Specific Technical Tasks:** - **Compound Synthesis**: Design and synthesize different bio-based phenolic antioxidants with varying molecular structures and functionalities - **Process Optimization**: Develop scalable synthetic routes suitable for industrial production - **Quality Control**: Establish analytical methods for purity assessment and quality control - **Fundamental Research**: Investigate structure-activity relationships and antioxidant mechanisms **Knowledge Transfer and Documentation:** - **Technical Reports**: Provide detailed synthesis protocols, analytical data, and performance results - **Training and Support**: Train Korean SME partner personnel in synthesis techniques and analytical methods - **Intellectual Property**: Lead patent applications for novel compounds and synthetic processes - **Scientific Publications**: Prepare peer-reviewed publications to disseminate research findings **Project Management Contributions:** - **Technical Coordination**: Lead technical aspects of the project and coordinate research activities - **Risk Management**: Identify and mitigate technical risks related to synthesis and performance - **Quality Assurance**: Ensure all research meets international standards and regulatory requirements.
Expected R&R of Korean SME Partner	**Korean SME Partner Responsibilities and Contributions:** **Industrial Application Expertise:** - **Market Requirements Definition**: Provide detailed specifications for tire and other rubber applications, including performance requirements, processing conditions, and cost targets - **Real-World Validation**: Conduct testing under actual industrial processing conditions using production-scale equipment - **Application Development**: Develop optimized tire compound formulations incorporating bio-based antioxidants - **Performance Benchmarking**: Compare new antioxidants against current 6PPD-based formulations in real rubber applications **Technical Contributions:** - **Rubber Compound Formulation**: Expertise in rubber compound design and optimization - **Industrial Processing**: Knowledge of large-scale rubber mixing, extrusion, and vulcanization processes - **Quality Control Systems**: Implementation of industrial quality control procedures for new antioxidants - **Field Testing**: Coordination of tire testing under real driving conditions and environmental exposures

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	Resource Commitments: - **Personnel**: Technical staff for compound development, processing trials, and performance evaluation - **Production Facilities**: Access to industrial mixing equipment, tire building facilities, and testing laboratories - **Testing Infrastructure**: Rubber testing equipment, road testing capabilities, and performance evaluation systems - **Market Intelligence**: Knowledge of customer requirements, regulatory landscape, and competitive positioning **Development and Testing Activities:** - **Compound Optimization**: Develop rubber formulations that maximize performance of bio-based antioxidants - **Processing Validation**: Validate processability and manufacturing feasibility at industrial scale - **Performance Testing**: Conduct comprehensive tire performance testing including durability, safety, and environmental resistance - **Demonstrator Production**: Manufacture prototype tires incorporating bio-based antioxidants for validation testing **Commercialization Support:** - **Market Analysis**: Assess commercial viability and market potential for bio-based antioxidant technology - **Customer Engagement**: Interface with tire manufacturers and end-users for feedback and validation - **Regulatory Support**: Assist with regulatory approvals and compliance requirements in target markets - **Supply Chain Development**: Establish supply chain relationships for raw materials and distribution **Collaboration Efforts:** - **Technology Transfer**: Facilitate transfer of laboratory-scale processes to industrial production - **Joint Problem Solving**: Work collaboratively to address technical challenges and optimize performance - **IP Development**: Contribute to joint patent applications and intellectual property protection - **Knowledge Sharing**: Share industrial expertise and market knowledge to guide research priorities **Scale-up and Implementation:** - **Pilot Production**: Scale up synthesis processes to pilot plant level (>10 kg batches) - **Cost Optimization**: Develop cost-effective production processes suitable for commercial implementation - **Quality Systems**: Implement quality management systems for consistent product quality - **Market Introduction**: Lead market introduction and customer adoption activities

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements:** **Novel Bio-Based Antioxidant Platform:** - Development of 3-5 commercially viable bio-based phenolic antioxidants with performance equivalent to or superior to 6PPD - Establishment of structure-activity relationships enabling rational design of future antioxidants - Scalable synthetic processes suitable for industrial production at competitive costs - Comprehensive performance database for various rubber applications **Environmental Benefits:** - Reduction of aquatic toxicity by >90% compared to 6PPD-Q transformation products - Enhanced biodegradability of antioxidant transformation products - Carbon footprint reduction of >30% through use of renewable feedstocks - Elimination of persistent toxic compounds from tire wear particles **Technical Innovations:** - Novel synthetic methodologies for bio-based antioxidant production - Advanced analytical methods for antioxidant performance evaluation - Optimized rubber compound formulations for sustainable tire applications - Predictive models for antioxidant performance in rubber systems **Commercialization Potential:** **Market Opportunity:** - Global tire antioxidant market valued at >$500 million annually - Growing demand for sustainable tire materials driven by environmental regulations - Premium pricing potential for environmentally friendly alternatives - First-mover advantage in bio-based tire antioxidant market **Business Impact:** - New revenue streams for both research institute and Korean SME partner - Enhanced competitive positioning in sustainable rubber technology - Licensing opportunities for developed technology and intellectual property - Potential for expansion into other rubber applications beyond tires **Industrial Transformation:** - Acceleration of sustainable rubber industry development - Reduction of rubber industry environmental footprint - Enhanced reputation and market positioning for participating companies - Contribution to circular economy principles in rubber industry **Broader Industrial and Economic Impact:** **Regulatory Compliance:** - Proactive response to anticipated 6PPD regulations - Reduced regulatory risk for rubber manufacturers - Enhanced ability to meet environmental standards and certifications - Improved corporate sustainability metrics **Supply Chain Benefits:** - Reduced dependence on petroleum-based antioxidants - Utilization of agricultural waste streams for value-added products - Enhanced supply chain sustainability and resilience - Support for bio-economy development **Knowledge Economy Contribution:** - Advanced scientific knowledge in bio-based materials chemistry - Enhanced research capabilities in sustainable rubber technology - Strengthened international research collaboration

	- Training of next-generation scientists and engineers **Societal Impact:** - Reduced environmental contamination from tire wear particles - Protection of aquatic ecosystems and biodiversity - Contribution to sustainable transportation systems - Enhanced public health through reduced exposure to toxic compounds **Long-term Strategic Impact:** - Positioning of Korean rubber industry as global leader in sustainable technology - Development of exportable clean technology solutions - Contribution to national sustainability and environmental goals - Foundation for future innovations in bio-based materials

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization